

Meeting Outline:

- Progress Update
 - Without intervention:
 - Have pipeline for looping through documents, generating summaries, predicting bias, storing into .csv file
 - Ran BART & T5 - working on GPT2, GPT Neo, Llama 2
 - GPT2 - better, but GPT Neo not so great
 - Prompt intervention:
 - Same pipeline as ^
 - Ran BART & T5 - working on GPT2, GPT Neo, Llama 2
 - SMC steering - not BART, T5
 - Position ID's are key issue
 - can resolve by not passing in position ids (which is only needed for caching)
 - main issue is that SMC steering requires Causal LM model, whereas BART's summarization is a ConditionalGeneration LM model -> incompatible
 - SMC steering wants autoregressive models like GPT and llama
 - Can discuss this as caveat
 - method
 - Twist method, which scales particles' weights with a factor at each timestep (higher is more probable) to indicate whether the particle is on the right track. Undoes the scaling after each timestep.
 - scaled by probability of target bias predicted by the bias model
 - pipeline
 - ^ Have code ready to be pushed to GH for multiple summaries, for each bias
 - Ready for GPT2
 - Need to run for GPT2, GPT Neo, Llama 2
 - ***
 - Datasets
 - Training data for bias model
 - stratified sampled 2500 articles from (BASIL + EMNLP paper)
 - equal numbers of each labeled articles (~800/bias)
 - Evaluation dataset for political bias fidelity scores
 - BASIL
 - Alternate steering approaches:
 - Activation
 - Weights
 - Attention
 - ^ Current focus = "Activation Addition"
 - <https://arxiv.org/pdf/2308.10248.pdf>
 - POC: Ben
 - fine-tuning the model itself with biased data

- out of scope and already covered by previous work
 - Side note:
 - can frame as
 - prompt-level intervention
 - decoding-level intervention (smc steering)
 - activation-level intervention (activation addition)
- Progress report
 - Finish by Sunday night
 - Goal - 6/8 pages total for final paper - ~2-3 more pages for progress - (this submission should be ~5 pgs.)?
 - Introduction (1.5 columns)
 - elaborate a bit more
 - preview the different levels of intervention
 - @Ben
 - Related works (1.5 columns)
 - related alternative steering approaches
 - @Ellie
 - Methodology (2 pages)
 - clean up “Evaluation” section from proposal
 - Dataset
 - @Ellie
 - bias model training details
 - @Reinaldo
 - PBI/PBF score detail (it has been modified from proposal so need to update)
 - Ellie - normalization [0,1]
 - Changed to be on increasing rather than decreasing scale
 - Changed infidelity to indicator function - either same or different (not *how* different, necessarily)
 - @Ellie
 - each intervention methodology
 - Baseline (no intervention)
 - @Rebecca
 - Prompt intervention
 - @Rebecca
 - SMC steering
 - @Ellie
 - Activation addition
 - @Ben
 - Faster inference
 - @Ben
 - Results & Discussion (1.5 pages)
 - figure/table of the results
 - can put placeholders for the progress report

- half page??
 - talk about limitation (so far)
 - @Ellie - Talk about SMC steering limitations
 - compare results
 - placeholder for now since we don't have all the results
- Conclusion (half a column)
 - Brief recap of results / interpretation
 - Implications for future studies
 - Placeholder for now since the project is not done